CUSTOMER CHURN ANALYSIS DOCUMENTATION

Objective

The objective of this project is to analyse customer churn behaviour on a digital platform using key behavioural and demographic metrics. The analysis aims to identify which factors—such as login device, satisfaction score, tenure, coupon usage, order activity, app usage time, and gender—are most strongly associated with customer churn. The insights will help the business design targeted retention strategies and reduce customer attrition.

Problem Statement

A digital commerce platform is experiencing a high level of customer churn, resulting in reduced repeat purchases, lower customer retention, and a negative impact on long-term revenue growth. Although the platform has a strong customer base and order volume, management lacks clear visibility into the key reasons why customers stop purchasing or disengage from the platform.

Several business challenges currently exist:

- There is no clear understanding of which customer segments contribute the highest churn, especially across device usage, gender, city tier, payment mode, tenure group, and order frequency.
- The business cannot identify whether new customers or long-term customers are more likely to churn.
- The relationship between customer satisfaction score and churn is not clearly measured.
- It is unclear whether coupon usage improves retention or only temporarily delays churn.
- The platform does not know how inactivity, such as days since last order, affects customer churn risk.
- Customer engagement metrics such as hours spent on the app are not being used to understand why some highly active users still leave.
- High-value product categories such as Mobile Phones, Accessories, and Fashion show significant customer loss, but the reasons are not clearly identified.
- Customers using certain payment methods, especially Cash on Delivery, appear to churn more frequently, but this pattern has not been analyzed in detail.
- Distance from warehouse locations may influence delivery speed and customer satisfaction, but its impact on churn is not currently tracked.
- Without proper churn insights, the business cannot create targeted retention campaigns, personalized offers, loyalty programs, or customer recovery strategies.

To solve these challenges, this project uses Power BI to analyze customer churn across multiple dimensions, including customer demographics, satisfaction score, payment preferences, coupon usage, order behavior, inactivity period, engagement level, and tenure group. The goal is to identify the key drivers of churn and provide actionable recommendations to improve customer retention, increase repeat purchases, and strengthen long-term business growth.

Attribute (Column /Features) Details:

The dataset consists of customer-level information capturing demographic, behavioural, and transactional attributes. Below is a description of each column.

Column Name	Description
CustomerID	Unique identifier for each customer
Churn	Binary flag: 1 if customer has churned, 0 if still active
Tenure	Number of months the customer has been with the platform
PreferredLoginDevice	Device used most often to access the platform (e.g., Mobile, Tablet, Desktop)
CityTier	Tier classification of the customer's city (1, 2, or 3, where Tier 1 = most metropolitan)
WarehouseToHome	Distance (in km) from the nearest warehouse/delivery hub to the customer's home
PreferredPaymentMode	Customer's most frequently used payment method (e.g., Credit Card, Debit Card, UPI, COD)
Gender	Customer's gender (Male / Female / Other)
HourSpendOnApp	Average number of hours per week spent on the mobile app / platform
NumberOfDeviceRegistered	Total number of devices (mobile, tablet, desktop) registered to the customer's account
PreferedOrderCat	Customer's most preferred order category (e.g., Technology, Fashion, Grocery, etc.)
SatisfactionScore	Customer-reported satisfaction score (typically 1 = very dissatisfied to 5 = very satisfied)
MaritalStatus	Marital status (e.g., Single, Married, Divorced)
NumberOfAddress	Number of different delivery addresses associated with the customer's account
Complain	Binary flag: 1 if the customer has filed a complaint in the last year, else 0
OrderAmountHikeFromlastYear	Percentage increase or decrease in order amount compared to the previous year (can be negative)
CouponUsed	Binary flag: 1 if the customer has ever used a coupon, else 0
OrderCount	Total number of orders placed by the customer during their lifetime
DaySinceLastOrder	Number of days since the customer's most recent order
CashbackAmount	Total cashback amount (in local currency) earned by the customer

Tools & Technologies

- **MySql:** Data cleaning,Transformation
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

Cleaning & Transformation:

1. Handle Missing Values

- **Numeric columns** (Tenure, WarehouseToHome, OrderAmountHikeFromlastYear, DaySinceLastOrder, HourSpendOnApp, CouponUsed, OrderCount):
  Replace NULL with the **mean** value of that column .
- **Categorical columns** (Gender, PreferredLoginDevice, PreferredPaymentMode, PreferedOrderCat, MaritalStatus):
  Replace NULL with the **mode** (most frequent value).
- **SatisfactionScore** (1–5 scale):
  Fill NULL with median (3) or mode.

2. Standardise Text Categories

- **PreferredLoginDevice**:
  Convert 'Phone' → 'Mobile Phone' (as already done).
- **PreferredPaymentMode**:
  Unify synonyms: 'CC' → 'Credit Card', 'COD' → 'Cash on Delivery', keep 'Debit Card', 'E wallet', 'UPI' as is.
- **PreferedOrderCat**:
  Ensure consistent capitalisation (e.g., 'Laptop & Accessory', 'Mobile', 'Fashion', etc.). Possibly map 'Mobile' to 'Mobile Phone' if needed.
- **MaritalStatus**:
  Standardise to 'Single', 'Married', 'Divorced' (no typos).

3. Fix Logical Inconsistencies

- **Tenure = 0 but OrderAmountHikeFromlastYear > 0**
  → Set Tenure to 1 (new customers shouldn't have a hike from last year).
  Tenure = 0 but OrderCount > 0
  → Could remain 0 (first order in same month). No change.
- **HourSpendOnApp = 0**
  → Replace with median value (e.g., 2) – more realistic than forcing 3.
- **WarehouseToHome = 0 or negative**
  → Treat as missing and replace with median.

4. Treat Outliers

- **WarehouseToHome**:
  Values above, say, 100 are unrealistic (distance in km/miles). Cap at 100 (or use IQR method).
  Original had 126, 127, etc. – likely data entry errors.
- **Tenure**:
  Cap at 60 months (5 years) – anything higher is improbable.
- **CashbackAmount**:
  If negative, set to 0.
- **OrderAmountHikeFromlastYear**:
  Extreme values (e.g., >100) could be capped at 100 or treated as outliers.

5. Ensure Correct Data Types & Ranges

- **Churn** and **Complain** should be BIT (0 or 1) – no action needed.
- **SatisfactionScore** must be between 1 and 5. Any value outside → set to nearest bound or median.
- **CityTier** should be 1, 2, or 3. Any other → set to mode (1).
- **NumberOfDeviceRegistered** – reasonable range (1–5).

6. Final Consistency Check

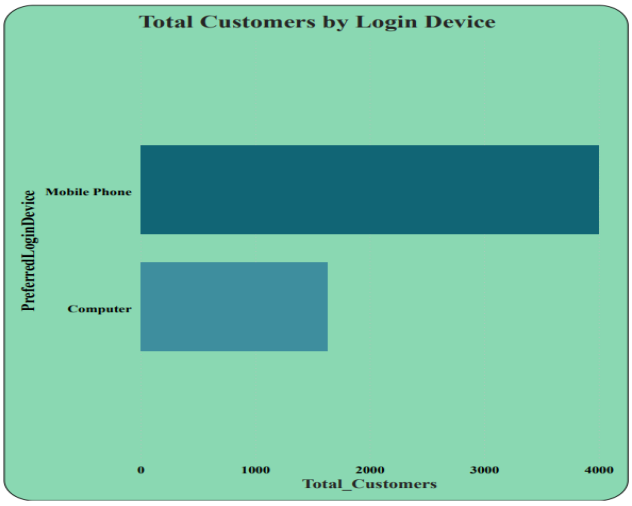
- Verify no NULL values remain in key columns.
- Check that Tenure is not negative.
- Confirm that DaySinceLastOrder is non-negative (0 = same day).

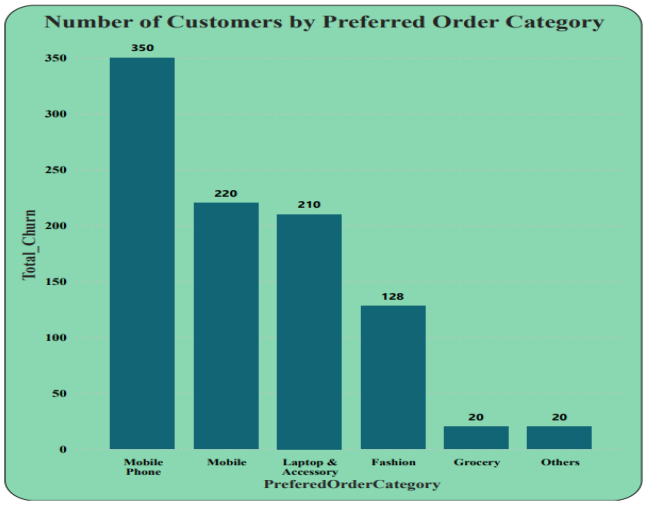
Data Transformation in Power BI

The following transformations were applied in Power Query / Power BI:

1. **Churn Rate Calculation**
   Churn Rate = SUM(Total_Churn) / SUM(Total_Customers)
   Shown as a percentage in visuals.
2. **Tenure Grouping**
   Tenure (in months) was binned into groups (e.g., 0–6, 6–12, 12–18, 18–24, 24+) to analyse churn by customer lifecycle stage.
3. **Days Since Last Order Binning**
   Created bins: 0–30 days, 31–60 days, 61–90 days, 90+ days to show how recency affects churn.

Visualizations Insights:



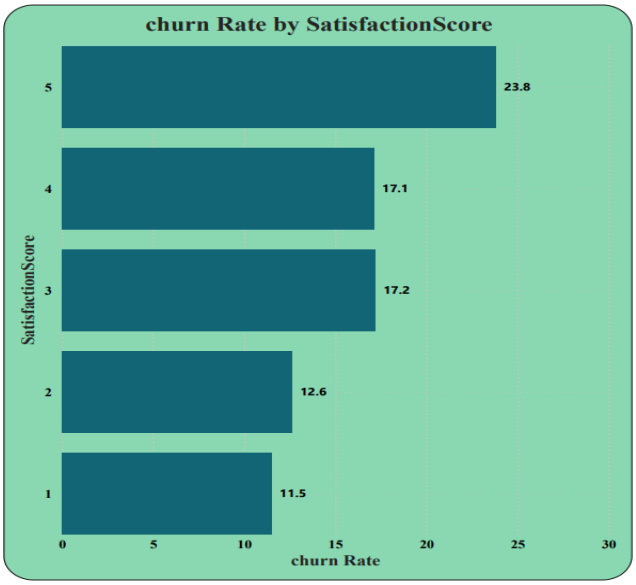


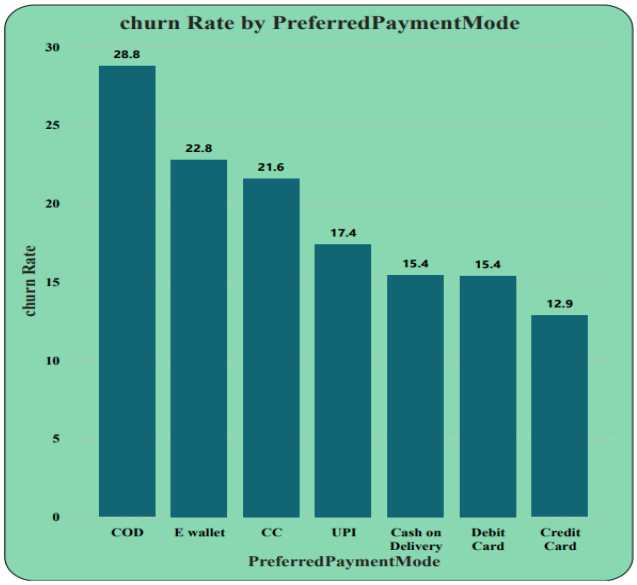
Insights:

- Mobile Phone users dominate the customer base, showing that the platform is heavily dependent on mobile traffic.
- The gap between Mobile Phone users and Computer users is extremely large.
- Mobile Phone category contributes the highest churn count among all product categories.
- Fashion and Accessories also show major customer loss despite being popular categories.
- Customers may be leaving due to pricing, lack of offers, poor delivery experience, or better competitor options.

Recommendations:

- Invest in mobile-first customer experience improvements.
- Introduce product comparison tools and faster checkout for mobile shoppers.
- Improve recommendations for Fashion and Accessories categories to increase customer engagement.
- Use abandoned cart notifications and category-specific discounts to recover at-risk customers.
- Monitor competitor pricing in Mobile Phones and Accessories because these are highly price-sensitive categories.



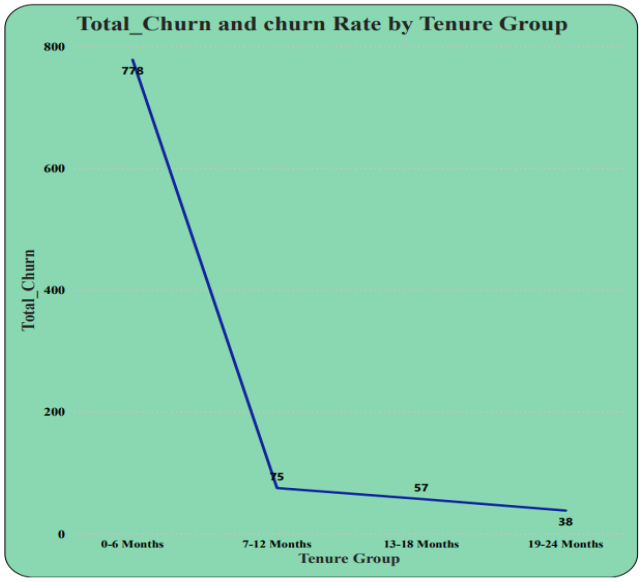


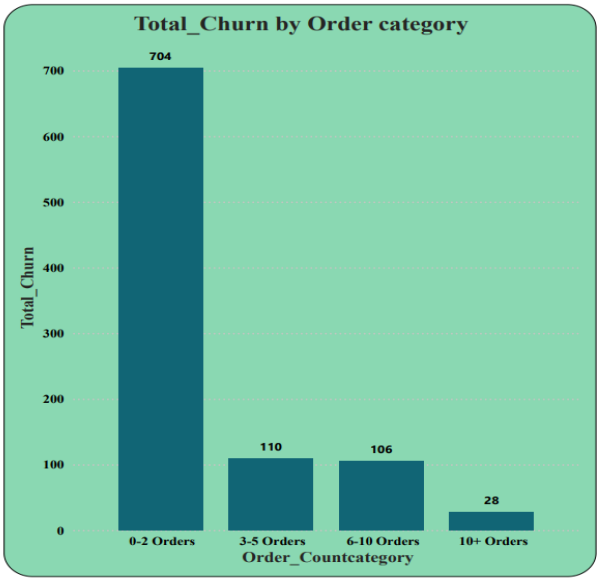
Insights:

- Customers using Cash on Delivery show the highest churn rate at around 28.8%.
- E-wallet users also show high churn at around 22.8%.
- Credit Card and UPI users have relatively lower churn rates.
- Customers using digital payment methods appear to be more loyal than customers using COD.
- COD customers may face issues such as delayed payments, order cancellations, or lower commitment to purchases.
- Customers with satisfaction score 2 have the highest churn rate at around 23.8%.
- Satisfaction score 1 also has a notable churn rate.
- Customers with satisfaction score 5 show the lowest churn rate.
- Lower customer satisfaction is strongly linked to higher customer churn.
- Even customers with moderate satisfaction scores are still leaving, which means the overall experience needs improvement.

Recommendations

- Encourage customers to use digital payments by providing cashback, discounts, or reward points.
- Reduce dependency on Cash on Delivery because it is associated with higher churn.
- Conduct customer feedback analysis for customers with low satisfaction scores.
- Improve post-purchase experience, product quality, customer support, and return process.
- Create a satisfaction improvement program for customers giving ratings between 1 and 3.
- Introduce follow-up surveys after delivery to identify customer pain points quickly.





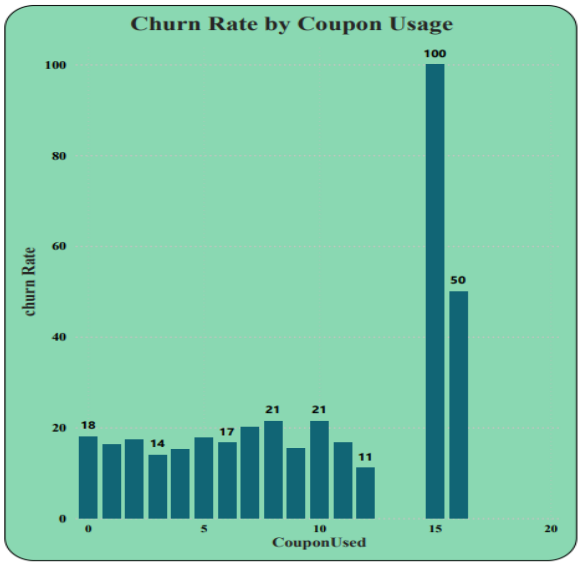
Insights:

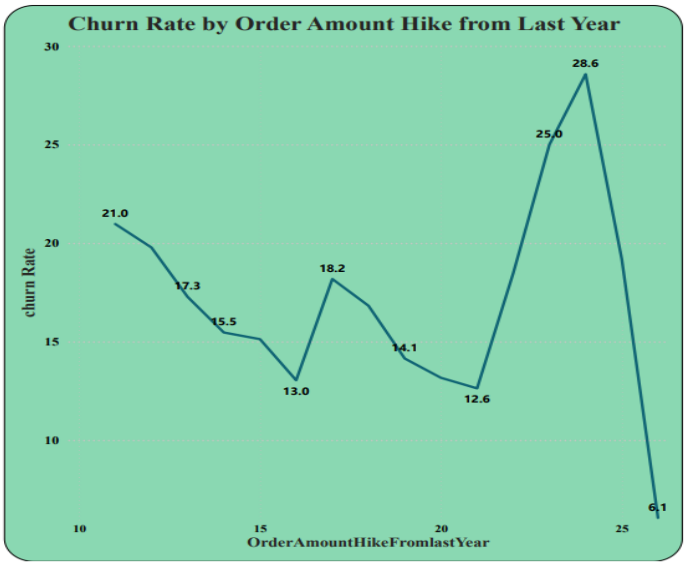
- Customers in the 0–6 months tenure group show the highest churn count at 778 customers.
- Churn reduces significantly for customers who stay longer on the platform.
- This indicates that the business is struggling to retain newly acquired customers.
- Customers who remain beyond the first 6 months are more likely to stay loyal.

- Customers with only 0–2 orders contribute the highest churn count at 704 customers.
- Churn decreases sharply as the number of orders increases.
- Customers with 10+ orders have the lowest churn count.
- Repeat purchase behavior strongly reduces churn risk.
- Customers who make very few purchases are more likely to leave permanently.

Recommendations

- Focus heavily on first-time customer retention.
- Launch onboarding campaigns, welcome discounts, and first-purchase rewards.
- Encourage second and third purchases quickly after the first order.
- Introduce loyalty programs for customers with fewer than 3 orders.
- Build retention journeys specifically for new customers during their first 6 months.
- Offer subscription models, repeat-order incentives, and membership benefits.





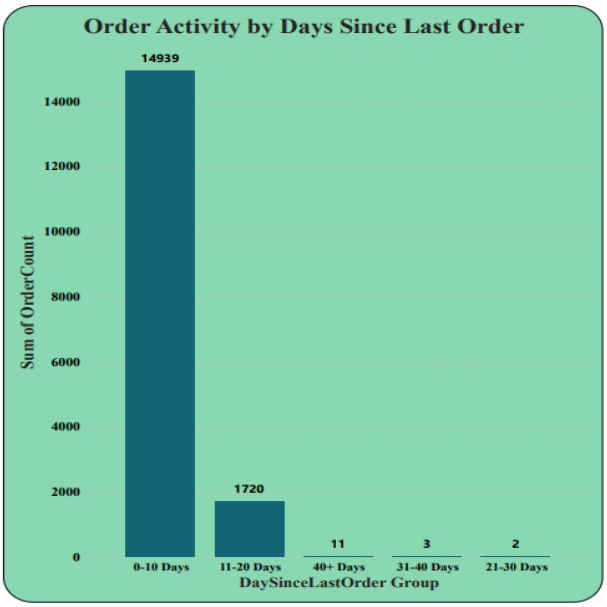
Key Insights:

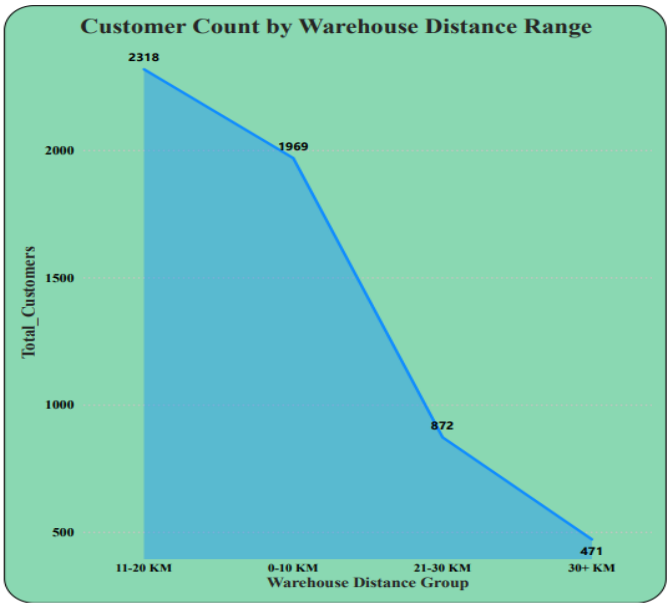
- Customers who do not use coupons show very high churn.
- Coupon usage appears to improve customer retention.
- Moderate coupon usage levels show lower churn compared to customers with no coupon usage.
- Customers respond positively to discounts and promotional campaigns.
- Customers experiencing a 20% order amount hike show the highest churn rate at around 28.6%.
- Sudden increases in spending are associated with higher customer loss.
- Customers may be sensitive to rising product prices, delivery charges, or hidden costs.
- Lower churn is observed when order amount increases are gradual and moderate.

Recommendations

- Use coupons strategically for customers showing early signs of churn.
- Offer discounts to inactive customers and low-order customers.
- Avoid aggressive price increases for returning customers.
- Provide loyalty discounts when product prices rise.

- Use targeted promotions for customers who reduced their order frequency.
- Create personalized coupon campaigns based on purchase history and category preference.



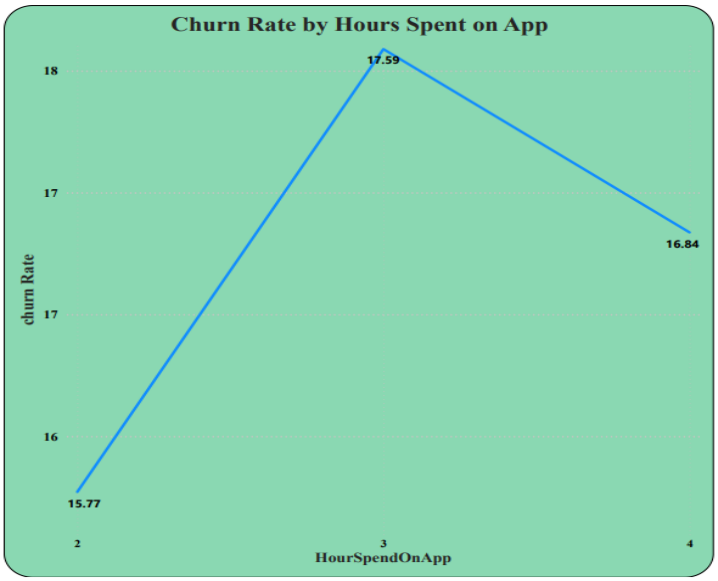


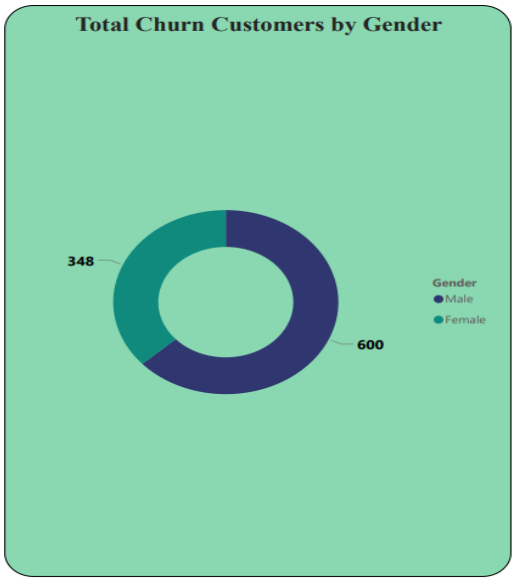
Key Insights

- Customers who ordered within the last 0–10 days contribute the highest order activity at 14,939 orders.
- Order activity drops sharply after 10 days.
- Customers inactive for more than 20 days are very unlikely to place future orders.
- Long inactivity periods are a strong indicator of churn risk.
- Most customers are located within 11–20 KM from the warehouse.
- Customers located 30+ KM away are fewer but may experience slower delivery times.
- Long delivery distance may reduce customer satisfaction and increase churn risk.
- Delivery convenience appears to be an important factor in customer retention.

Recommendations

- Identify customers inactive for more than 10 days and target them with reactivation campaigns.
- Send reminders, personalized offers, and recently viewed product recommendations.
- Improve delivery speed for customers located farther from warehouses.
- Consider adding more warehouse locations or local delivery partners.
- Offer faster delivery benefits for loyal customers in distant locations.



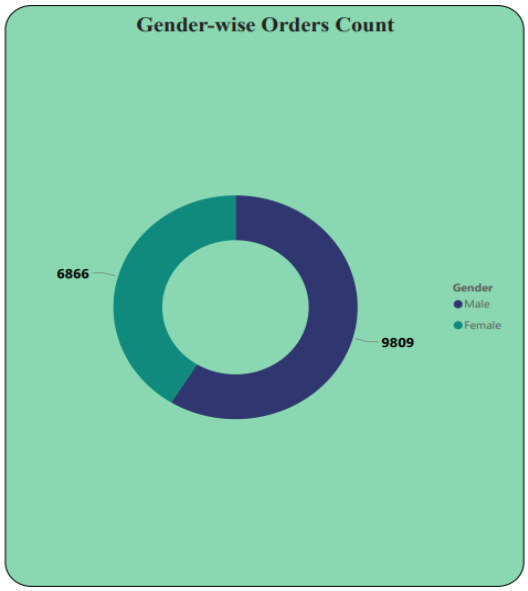


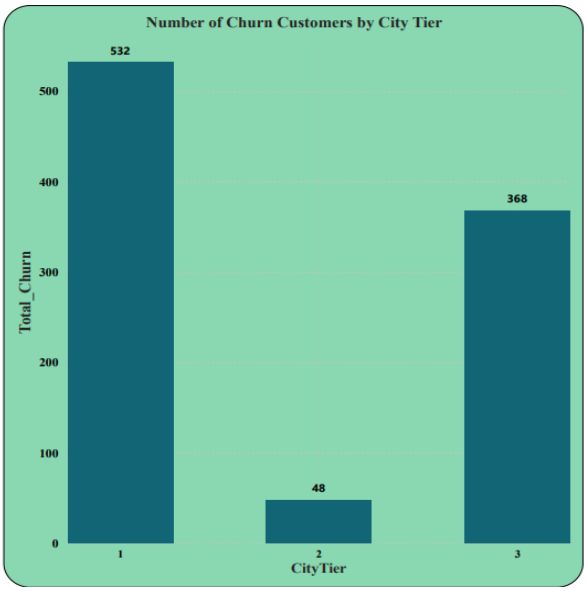
Key Insights

- Customers spending more time on the app show slightly higher churn rates.
- Customers spending 4 hours on the app have the highest churn rate.
- High app usage with high churn may indicate browsing without successful purchase completion.
- Customers may be facing issues such as pricing, checkout complexity, or poor product availability.
- Male customers contribute 600 churn customers.
- Female customers contribute 348 churn customers.

Recommendations

- Improve user journey and checkout flow for highly active app users.
- Study where customers drop off during the shopping process.





Key Insights

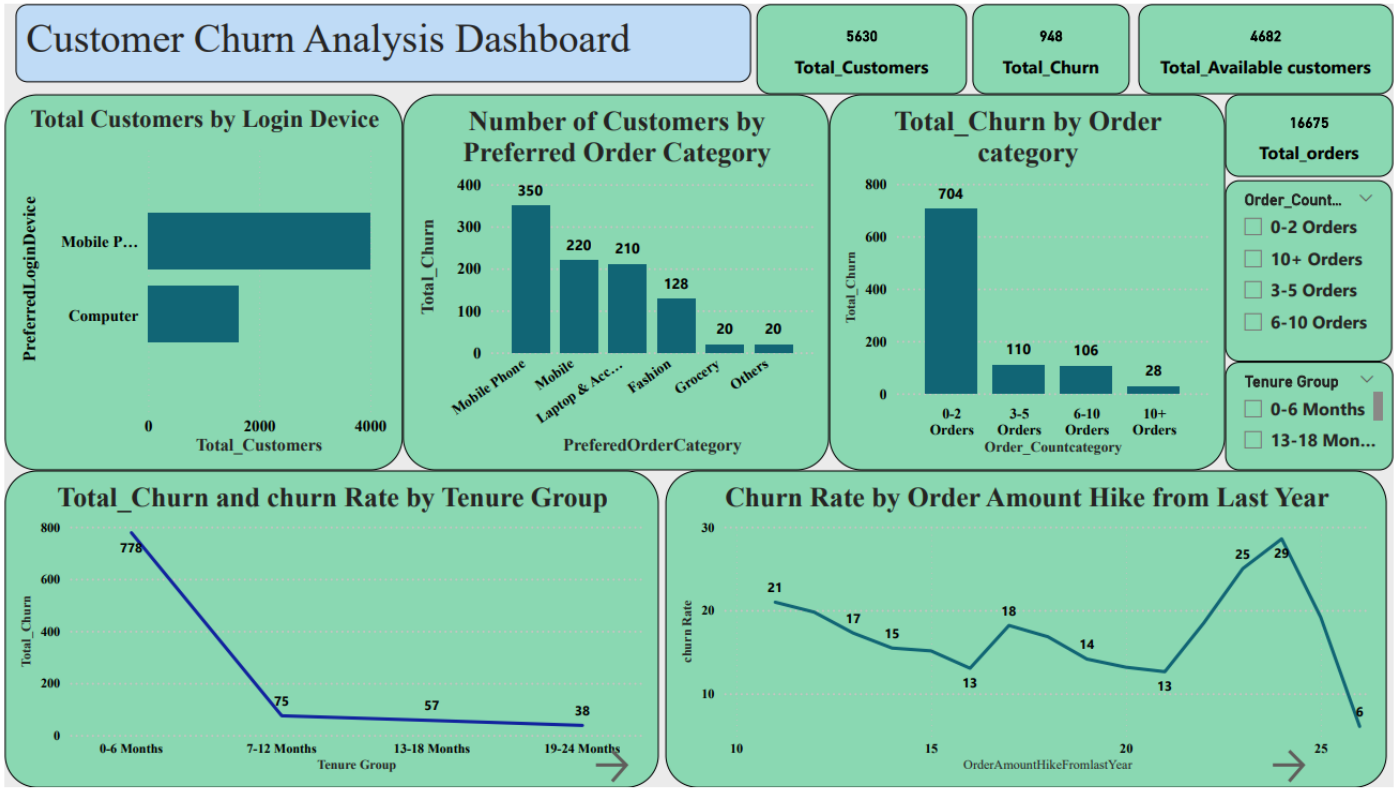
- Male customers show higher churn because they also likely represent a larger share of total orders.
- Tier 1 cities contribute the highest churn count at 532 customers.
- Tier 2 cities contribute 368 churn customers.

- Tier 3 cities have very low churn.
- High churn in Tier 1 cities suggests stronger competition, higher customer expectations, and more alternatives.

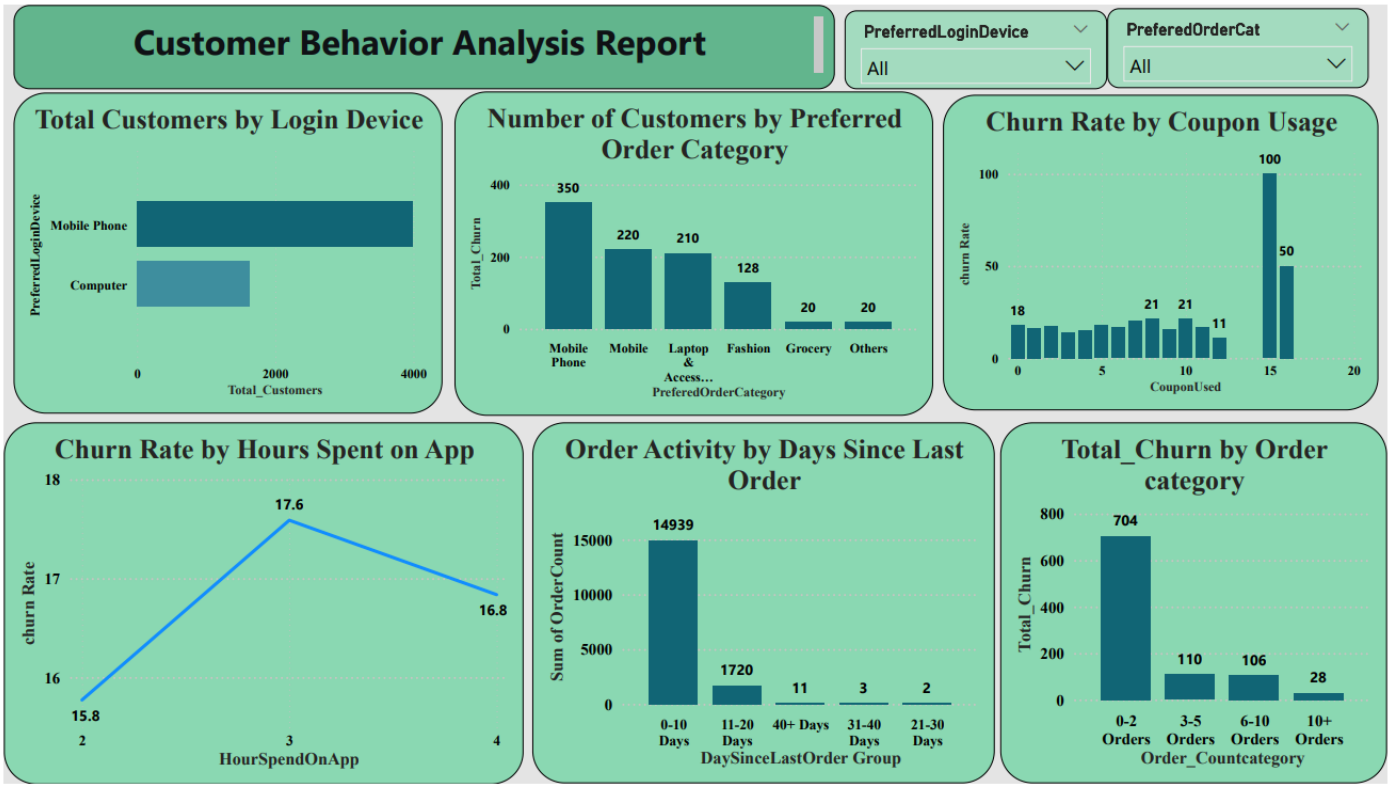
Recommendations

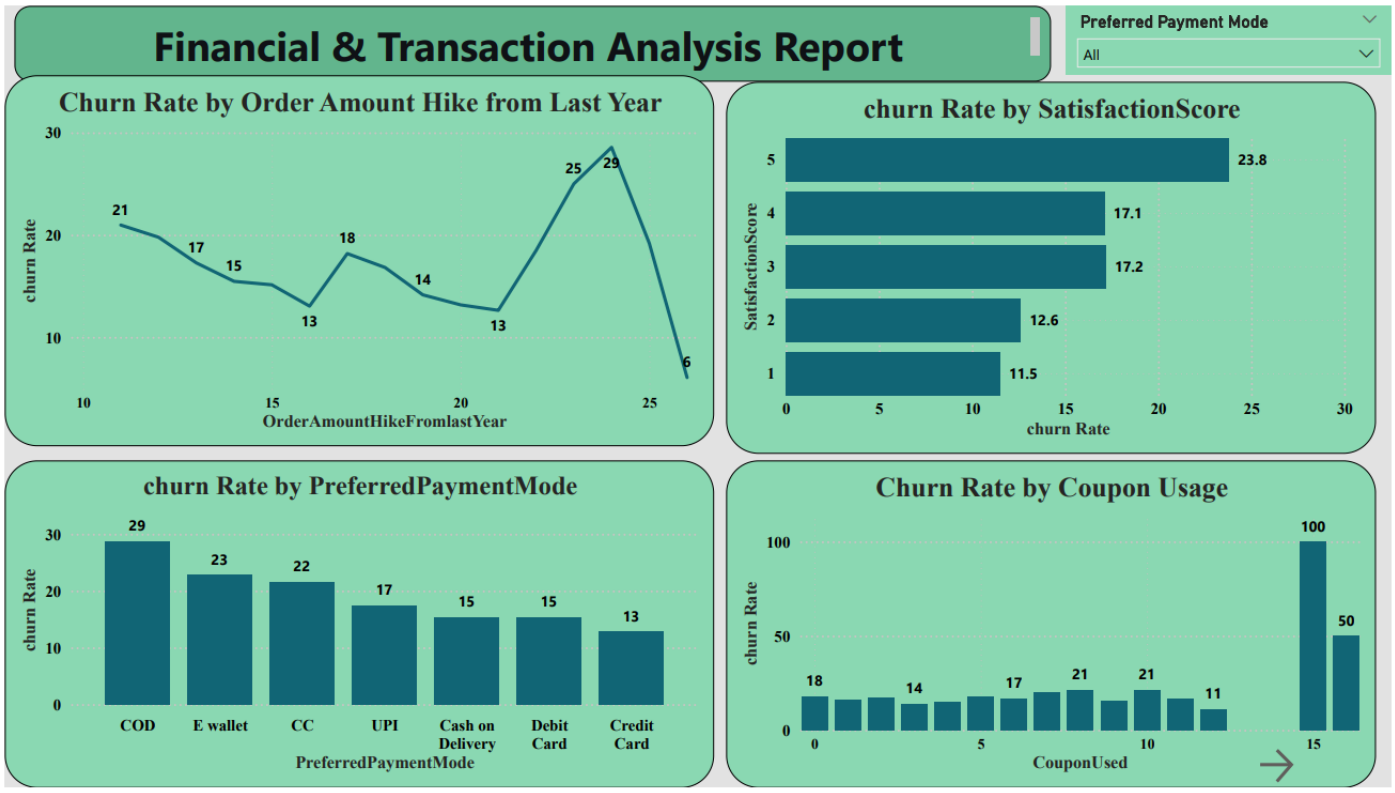
- Create separate retention strategies for male and female customers.
- Focus heavily on Tier 1 city customers because they represent the highest churn opportunity.
- Use premium offers, faster delivery, and better customer service in Tier 1 markets.
- Increase local marketing and personalized offers in Tier 2 cities to prevent future churn growth.

Final Dashboard And Reports:



Reports:





Conclusion

- The business has a major churn issue, with 948 customers leaving out of a total customer base of 5,630.
- Most churn is coming from new customers, low-order customers, and customers in their first 6 months.
- Mobile Phone users dominate the platform, making mobile experience the most critical business priority.

- High-churn product categories such as Mobile Phones, Accessories, and Fashion require stronger customer retention efforts.
- Cash on Delivery users and customers with low satisfaction scores are more likely to churn.
- Customers who use coupons, place repeat orders, and remain active within 10 days are more likely to stay loyal.
- Tier 1 cities contribute the highest churn volume, showing that competition and customer expectations are highest in these markets.
- Delivery distance, payment preferences, order frequency, and customer satisfaction all have a direct impact on churn.
- The company can significantly reduce churn by focusing on first-time customers, improving mobile experience, encouraging digital payments, offering personalized discounts, and building strong loyalty programs.
- If these improvements are implemented, the platform can improve retention, increase repeat purchases, and strengthen long-term revenue growth.